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**DRUG SUBSTITUTE IDENTIFICATION AND RISK
ANALYSIS IN THE PHARMACEUTICAL SUPPLY
CHAIN USING DATA-DRIVEN SIMILARITY AND
EXPLORATORY ANALYTICS**

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(Student signature)

A handwritten signature in black ink. It features a large, stylized capital 'F' that loops around and under itself. To the right of the main loop, the initials 'F.M.' are written in a smaller, cursive script.

Fatemeh Mashayekhahangarani

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EXECUTIVE SUMMARY

Overview and Problem Statement

In the context of the modern pharmaceutical system, drug shortages are not only a logistical failure but also an important threat to patients' safety and the stability of the healthcare system. Traditional approaches to managing drug supply chain shortages are often reactive, manual, and rely on fragmented data, but this project addresses the issue of information gaps in the supply chain by developing a data-driven system capable of identifying clinically valid alternative drugs through advanced semantic analysis and multidimensional risk mapping.

The Solution: A Hybrid Data-Driven Framework

The main core of this research is the transition from simple keyword matching to semantic text embeddings. By analyzing a dataset of 5,000 drug entries using the Sentence-Transformer (all-MiniLM-L6-v2) model, we developed a system that understands the medical meaning behind drug descriptions. Unlike tools which are using today's, this project used a hybrid approach that combined unstructured medical texts with structured treatment classes. This ensures that the proposed alternatives are not only linguistically similar, but also medically suitable for replacing each other, as demonstrated by the results obtained.

Key Indicators for Decision Support (SI & SRI)

To transform complex data into management actions, two dedicated key performance indicators (KPIs) were introduced:

- **Substitutability Index (SI):**
Measures the operational flexibility of a drug by quantifying its available alternatives.
- **Shortage Risk Index (SRI):**
Functions as a strategic warning system, flagging "Red Alert" drugs that have low substitutability and require prioritized procurement and safety stocks.

Findings and Validation

The model performance was validated using external datasets and the model's reliability in real-world scenarios was confirmed by achieving high Hit@k and Precision@k scores. The results showed that the integrated data obtained by combining text and clinical structures provided a more robust representation of drug relationships than traditional databases. Visualization through t-SNE clusters also confirmed that the model was able to successfully organize the drug space into specific intuitive and clinical categories.

Strategic Impact

This study shows a scalable Decision Support System (DSS) that moves healthcare organizations from reactive to proactive risk management. By implementing this framework, all stakeholders mentioned in part 3.1 can reduce time, and human errors in their decision-makings, and create a resilient supply chain that prioritizes patient care even during market fluctuations.

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

In today's world, drug shortages have gone from being a simple issue to a chronic problem. When a drug becomes scarce or out of stock, the primary response from suppliers is to find and replace it with another drug. In the past, this was done by manual search and over-reliance on humans, which resulted in slow substitution processes and even errors in selecting the replacement drug.

Today's substitution methods are mostly based on the active ingredient of the drug. This one-dimensional selection method ignores other fundamental factors such as chemical structure, mode of action, or side effects that are very important in determining the choice of a substitute drug. For companies which working on medicines, this type of selection can be dangerous because choosing the wrong substitute not only means reduced or no sales of the drug, but can also lead to legal liabilities, including financial and life-threatening losses to patients (Aronson et al., 2023a; Aronson et al., 2023b).

Although pharmaceutical supply chain problems, from raw material shortages to over-reliance on a particular manufacturer, have been well addressed in previous research (Andy and Andy, 2023), the main solution to this problem seems to lie in the way information is processed. Traditional ERP systems in pharmacies are mostly descriptive in nature, meaning they only have the ability to tell us whether a drug is available or not, but they are not intelligent enough to suggest what can replace a rare or scarce drug. By using advanced data-driven techniques, especially Text Embeddings and Structural Similarity, we can turn existing pharmaceutical data into a tool for decision-making (Kauffman et al., 2025). In this study, Kauffman shows that he and his colleagues were able to design and build a multi-dimensional similarity measurement system that helps pharmaceutical managers make quick, data-driven choices in emergency situations.

1.2 PROBLEM STATEMENT

Currently, the pharmaceutical supply chain is facing a serious crisis. While the problem of drug shortages is increasing day by day, there is little standard data-driven system that can fully automatically identify alternative drugs. In practice, pharmaceutical actors are forced to rely on manual searches or their previous experiences or experts in this field, which will lead to the Business Dilemma, one is the slow decision-making process in critical situations and the other is the lack of accuracy in systems that only match drugs based on the active ingredient of the drugs and ignore the complex relationships between the chemical structure, mechanism of action and therapeutic use of drugs.

In terms of management, a process based on trial and error is not only inefficient but can also carry high risks in various aspects. Choosing alternative drugs over systems that operate solely on the basis of the active ingredient can lead to tension in the supply chain, non-compliance with the regulations

and standards of organizations overseeing the drug supply process, as well as loss of trust among doctors, pharmacies, and hospitals.

Therefore, the main problem of this study is:

How can we integrate multiple textual data and chemical and therapeutic properties to create a highly accurate model, and how can we translate these analytical outputs into actionable business insights to mitigate risks from drug shortages?

As we have said, although there are some traditional databases for drug substitution that have been used in various centers, their function is mainly as static search tools that require human interpretation and manual reviews. The business value of our proposed system is that it has passed from the "information retrieval" stage to the "intelligent recommendation" stage and, by integrating multidimensional data, provides a Quantified Score that reduces the mental burden of managers and decreases the risk of human errors in critical situations. This automation not only speeds up the purchasing process, but also ensures that the substitution decisions made are aligned with both "therapeutic safety" and "supply chain efficiency". This creates a competitive advantage in the current market.

1.3 RESEARCH AIM AND OBJECTIVES

Research Aim:

The main point of this thesis is to develop an advanced, multidimensional framework for identifying alternative drugs by integrating textual data with drug characteristics. In this study, we have attempted to move beyond simple keyword matching to create a data-driven decision support tool that helps pharmaceutical executives mitigate the effects of drug shortages through high-accuracy similarity modeling.

Research Objectives:

1. Develop Semantic Representations:

To generate high-dimensional text embeddings from drug descriptions, capturing the semantic meaning of clinical and pharmaceutical narratives.

2. Encode Multimodal Structural Features:

To systematically encode structured categorical data, including therapeutic classes, chemical structures, and mechanisms of action.

3. Implement Dimensionality Reduction and Optimization:

To address high-dimensionality challenges using techniques like PCA and to determine the optimal cluster configurations (k-selection) through Systematic Optimization (Elbow and Silhouette methods).

4. Synthesize Feature Fusion:

To integrate textual and structured vectors into a unified similarity matrix using Cosine Similarity for robust substitute identification.

5. Map the Substitution Network:

To identify Top-k substitutes and visualize the drug landscape through clustering and network analysis to identify substitution "hubs".

6. Develop Managerial KPIs:

To construct the Substitutability Index (SI) and Shortage Risk Index (SRI) as concrete business metrics for evaluating supply chain vulnerability.

7. Validate via External Benchmarking:

To rigorously evaluate the model's performance using an external validation dataset through precision-based metrics.

1.4 RESEARCH QUESTIONS

This study seeks to answer the following basic questions:

RQ1: Technical integration

How does embedding text structures with drug properties such as therapeutic class, chemical, and mechanism of action improve the accuracy of alternative drug identification compared to traditional methods that use only one data source?

RQ2: Systematic optimisation

To what extent does hyperparameter optimisation, such as selecting the optimal number of clusters through elbow and silhouette methods, improve the reliability of drug grouping in the substitution network?

RQ3: Management risk assessment

How can the Substitutability Index (SI) and Shortage Risk Index (SRI) be used to categorize therapeutic classes based on their supply chain vulnerability and substitution potential?

RQ4: Practical support for decision-making

How can a data-driven model become a practical tool to support purchasing managers' decision-making to reduce human error and their intellectual burden during times of critical shortages?

1.5 SCOPE & SIGNIFICANCE

Scope

This study was conducted on a dataset called MID, which includes columns such as drug name, clinical description, therapeutic class, chemical structure, and mechanism of action. This study was deliberately limited to only the features mentioned in the data columns to test the performance of a multidimensional similarity model without relying on variables that may fluctuate, such as price or sales data. We also used an external validation dataset to ensure the performance of the model and test the accuracy of substituted drugs. Although this study does not involve financial modeling and/or models with instantaneous fluctuating variables, it can provide the building blocks of a fundamental analytical architecture for advanced planning in purchasing and warehousing.

Significance

The importance of this study is the potential of transformation and how pharmaceutical supply chains respond to systemic drug shortages. As Aronson et al. (2023a) noted in their study, manual and traditional substitution models are a major high-risk bottleneck in the pharmaceutical supply chain, which also increases clinical and operational risks.

The practical achievements of this research include three main ones:

- **Decision Acceleration:**

By replacing slow manual search systems with an intelligent "Top-k" recommender system, this model reduces response time in situations where a drug is becoming scarce or out of stock.

- **Multi-Dimensional Reliability:**

Unlike simple tools that only consider the active ingredient of the drug, the model proposed in this project integrates "textual structures" with "drug structural features," ensuring that alternatives are therapeutically safe and feasible.

- **Strategic Risk Management:**

By introducing SI and SRI indicators, this study provides managers with predictive insight to identify vulnerable drug categories before shortages occur, enabling the drug supply chain system to shift from a passive to a pro-active mode.

Ultimately, this research serves as a scalable prototype for smart supply chains, bridging the gap between raw data science and real-world pharmaceutical resilience.

1.6 OVERVIEW OF ANALYTICAL APPROACH

This study follows a systematic, eight-step analytical pipeline designed to bridge the gap between raw pharmaceutical data and actionable business intelligence:

Step 1: Exploratory Data Analysis (EDA) & Chi-Square Testing

Beyond basic data cleaning and missing value analysis, this step includes Chi-Square tests to identify statistical dependencies between therapeutic classes and drug properties, providing an initial understanding of the data structure.

Step 2: Advanced Text Embeddings

Clinical drug descriptions are transformed into high-dimensional numerical vectors using embedding techniques. This allows the model to capture the semantic context of how a drug is described, rather than just matching keywords (Kauffman et al., 2025).

Step 3: Multi-Label Encoding of Structured Features

Categorical data such as Therapeutic Class, Chemical Structure, and Mechanism of Action are encoded into numerical formats. This method ensures that the "scientific DNA" of the drug is represented alongside its textual description.

Step 4: Dimensionality Reduction (PCA) & Feature Fusion

To handle the high-dimensionality of the combined data and remove "noise", and **Principal Component Analysis (PCA)** is applied. The refined textual and structured features are then fused into a single unified representation for each medicine.

Step 5: Systematic Optimization & Clustering

Instead of arbitrary grouping, we use the Elbow Method and Silhouette Scores to determine the optimal number of clusters (k). Drugs are then grouped using K-Means clustering to identify natural substitution families.

Step 6: Similarity Calculation & Network Mapping

Using Cosine Similarity, the model identifies the Top-k closest substitutes for every drug. These relationships are visualized through a Substitution Network, highlighting "hubs" where drugs are easily interchangeable.

Step 7: Robust Model Evaluation

In this step, the model's performance is rigorously tested against an external benchmark. We use technical metrics like Hit@k and Precision@k to measure how often the model correctly predicts a known substitute.

Step 8: Strategic Business Synthesis (SI & SRI)

Finally, two indicators are created:

- **SI (Substitutability Index)**
- **SRI (Shortage Risk Index)**

In the final stage, the analytical outputs are converted into two indices: the Substitutability Index (SI) and the Shortage Risk Index (SRI). These indicators allow drug supply chain managers to identify which drug categories are "at risk" and require proactive procurement strategies (Petersen et al., 2016).

This approach makes the analysis technically strong and also useful for real decision-making in the pharmaceutical supply chain.

1.7 STRUCTURE OF THE DISSERTATION

This dissertation is divided into nine parts.

- In first Chapter it introduces the research background, the central business problem of drug shortages, and the study's specific aims and questions.
- Chapter Two presents the literature review. This chapter brings together existing academic research on pharmaceutical supply chains, traditional substitution methods, and the role of machine learning in healthcare.
- Chapter Three establishes a connection between supply chain management needs and analytical capabilities and defines how data science can solve management problems.
- Chapter Four details the MID dataset and presents statistical findings, including missing-value analysis and Chi-Square tests to uncover hidden patterns in drug properties.

- Chapter Five provides a detailed theoretical framework of the selected models, explaining the choice of Sentence-BERT (SBERT) for text embeddings the logic behind structured feature encoding, and the mathematical basis for Cosine Similarity and K-Means clustering.
- In Chapter six, the actual implementation of the model is carried out. This chapter includes converting texts to vectors, reducing dimensions using PCA, systematic clustering optimization (Elbow/Silhouette), and building a replacement network.
- Chapter Seven, Validation & Evaluation. In this chapter rigorously tests the model's accuracy using the external dataset, calculating performance metrics like Hit@k and Precision to ensure reliability.
- Chapter Eight, in this chapter, technical outputs are transformed into strategic tools, particularly with a focus on visualizing the substitution network and calculating the SI (Substitution Index) and SRI (Shortage Risk Index).
- Finally, Chapter Nine summarises the project's contributions to the pharmaceutical industry, discusses limitations, and suggests future paths for intelligent supply chain systems.

CHAPTER 2

LITERATURE REVIEW

2.1 DRUG SHORTAGES: DEFINITIONS, CAUSES, IMPACTS

Today, the pharmaceutical supply chain has become extremely vulnerable. The problems are not just related to simple delays at the factories. Aronson et al. (2023a) argue that shortages occur when patients cannot get the medicine they need, but there is a larger risk called “single-point-of-failure.” This means that the entire system is often dependent on just one supplier or a specific way of working and this kind of dependency on a lone supplier or a rigid workflow creates a fragile system where a minor hiccup can halt all the process.

Also, in some studies conducted by Adak (2024) in the past, drug shortages are related to transportation or shipping problems, but there too the problem of Information Asymmetry is not considered. Information Asymmetry is not just a technical problem but a management problem in which managers responsible for purchasing drugs do not have enough clinical information to quickly find alternatives, which leads to “reactive” decisions, meaning that managers only take action when the drug is no longer available on the market. This lack of forecasting tools doesn't just disrupt the supply chain. It creates a tension between the pharmacy's need for inventory and the patient's need for safety. Without early proactive tools, managers are left with an unacceptable level of risk in the drug supply process.

2.2 MEDICINE SUBSTITUTION: CONCEPTS & CHALLENGES

At first, replacing a drug with other one may seem simple, but this process is much more complex than it seems. The real challenge lies in this simplicity: identifying which alternative is truly effective. This means carefully assessing whether the alternative is clinically safe for the patient and is practically feasible in the system, a process that is currently time-consuming and slow in current systems.

In general, in the traditional manual and semi-automatic substitution systems currently available in the pharmaceutical market, we are dealing with two levels of substitution. First, generic substitution, which seems simple because it contains the same active ingredient, but even this simple substitution may carry risks because differences in chemical structures or excipients can significantly affect patient treatment outcomes. The second level of substitution is therapeutic substitution. At this level, the manager must find a drug with a different active ingredient that still achieves the same clinical goal as the previous one, which requires in-depth pharmaceutical knowledge of the managers, which most drug supply chain managers or purchasing officials lack:

- **The chemical structure and mechanism** of action of the drug to ensure that the drug works in the same way as the original drug
- **Therapeutic class and side effects** of medications to ensure patient health
- **Drug interactions** to prevent new medical complications

In the current tools available on the market, whether they are traditional manual or semi-automated, the tension between the need for rapid replacement and the demand to maintain patient health is very high. Therefore, the move towards the use of data-driven tools and machine learning is not only a technical upgrade but also a business necessity. By using Similarity Models that combine structured data, we can choose the best alternative instead of “guessing” through “mathematical calculations”, which greatly reduces the operational and medical risks inherent in current methods.

2.3 DATA-DRIVEN APPROACHES IN HEALTHCARE & SUPPLY CHAIN

The shift towards data-driven approaches to drug alternative selection is not just a trend, but a response to the growing inefficiency or even inefficiency of existing, traditional management systems. With drug shortages on the rise for various reasons, relying on emotions or empirical evidence is no longer a viable business strategy, further highlighting the need for data-driven decision-making and transforming raw data into classified information for planning a proactive automated system.

The following analysis attempts to take a data-driven approach beyond the mere collection of data. It requires a systematic flow of data refinement, pattern extraction through exploratory analysis, and building models that can truly support a manager’s choices. Currently, many healthcare systems are “raw data rich but lack systematic insight.” This means that managers have the raw information they need to make effective decisions but lack modern tools such as data clustering models to help managers, physicians, and pharmacists make decisions, which is the goal of this project.

Today, the real value of machine learning methods in all fields is their ability to handle high-dimensional data that the human brain cannot process quickly and accurately. For example, in the field of pharmaceutical activities, while traditional systems may only look at the name of the drug, advanced techniques such as text embedding and similarity search allow us to understand a deeper relationship between the chemical structure of the drug and its clinical use.

These tools are now a necessity because:

- **The scale of data is too large:** Manual sorting of thousands of drug interactions is impossible.
- **Speed is a competitive advantage:** In a shortage, the faster a manager finds a safe substitute, the lower the operational risk.
- **Human error is costly:** In healthcare, a mistake in substitution is not just a financial loss; it is a safety risk.

2.4 TEXT EMBEDDINGS, SIMILARITY METHODS & ML FOR SUBSTITUTION

The real success of a drug shortage management system lies in how it transforms complex medical texts into actionable data. While traditional databases rely solely on keyword matching, text embedding and the use of modern machine learning systems is a superior technical approach

because semantic meaning encompasses all the characteristics of drugs. As Kaufman (2025) suggests in his research, converting text into vectors helps the computer understand the true meaning behind the words. This is crucial for projects like the one we are about to discuss, as a simple label like “painkiller” doesn’t tell the whole story. Detailed descriptions of the “action” or “treatment” of a drug contain small but important details about how the drug works. Using these numerical vectors, the computer can see these hidden details that a simple category name would miss and select drug alternatives much more accurately.

This project specifically focuses on cosine similarity as a key metric for the substitution model. From a management perspective, it is not just a mathematical formula but an objective ranking tool. Instead of a pharmacist relying on memory or generic books, the similarity score metric can easily measure the degree of similarity between two drugs. This reduces the psychological burden of human error on decision makers and provides a level of consistency that manual systems lack.

2.5 SUMMARY OF LITERATURE GAPS

After reviewing the existing literature and research, several fundamental gaps in the pharmaceutical supply chain domain are visible, which justify the need for this study. While many researchers have examined drug shortages and their analysis separately, the approach of this project is that the missing link in the automated drug substitution system is an integrated system that links clinical data with management decision-making.

First gap:

The first gap that was evident in previous studies and reviews is that most studies are very fragmented. Some, such as Aronson et al. (2023a), focus only on the causes of shortages, while others, such as Andy and Andy (2023), focus only on the act of substitution. This kind of one-dimensional approach is a major business risk. A manager cannot solve a drug shortage without an immediate and reliable substitution tool. The research being conducted here seeks to fill this existing gap by considering shortage management and substitution identification as an integrated process.

Second gap:

The second tangible gap is a technological and evaluation gap. Current manual systems are often outdated and limited to examining “active ingredients.” Even when machine learning is mentioned in other studies, it is rarely used to combine “textual embeddings” with “structured chemical features.” The ongoing project believes that without standardized metrics like Hit@k or Precision@k, we cannot truly trust the reliability of a model in a high-risk field like healthcare.

Third gap:

Finally, the lack of practical “actionability” is another visible gap in the current pharmaceutical supply chain. Much of the research remains theoretical. As Ayer (2025) points out, data-driven systems must support real-world decisions. Here, we believe that simply showing “similarity” is not enough for a business manager. That is why we will introduce two new indicators: the Substitution Index (SI) and the Substitution Risk Index (SRI).

By filling these gaps, this study goes beyond a mere descriptive article. It provides a scientific, data-driven, and most importantly, practical tool that transforms complex raw data into transparent decisions for the pharmaceutical supply chain that can practically help managers and activists in the healthcare and pharmaceutical sector to predict and act quickly and with high confidence before a shortage of a specific drug occurs, while maintaining clinical safety as well as patient safety with their choices.

CHAPTER 3

BUSINESS–ANALYTICS INTEGRATION

3.1 STAKEHOLDERS & DECISION CONTEXT

In the pharmaceutical supply chain, identifying stakeholders is only the first step. To ensure the usability of this project, we must first understand how data insights are translated into specific actions for each group. Based on the analysis conducted, the value of this model lies in its ability to bridge the **information gap** between the following key stakeholders:

Pharmacists and Frontline Clinicians

The main challenge for pharmacists is time pressure and the risk of human error during drug shortages. The results of our project can be applied directly to this group. One of the parameters discussed in this project is the similarity score, which provides an objective list of alternatives that, instead of searching through manual records, can be used to quickly identify drugs with the highest clinical “closeness”, speeding up the selection of alternatives without compromising patient safety.

Physicians and Medical Professionals

Doctors are concerned about the clinical consequences and high risk of choosing alternative drugs. This is where the importance of using the Substitution Risk Index (SRI) comes into play. Knowing that a drug has a high SRI is a vital warning to doctors. This number tells them that substitution is not a simple process and requires close medical supervision. Ultimately, the findings of this project turn a simple data point into a vital safety tool that will be very useful to doctors.

Supply Chain and Procurement Managers

From a business perspective, managers need to shift from a reactive to a proactive planning strategy. The Substitution Index (SI) is useful for this group of managers in strategic inventory management. By identifying drugs that have a low SI (meaning they are difficult to replace), managers can prioritize these items in their procurement contracts and contingency stocks to avoid complete shortages of those drug items.

Health Authorities and Policy Makers

At the macro level, healthcare managers are responsible for the stability of the system. Using the findings of this project and understanding which drug classes are statistically related and which are disjoint, they can easily identify vulnerable drug clusters and design more flexible distribution policies to protect the entire healthcare system from sudden market shocks.

Finally, the model presented in this project not just as a technical tool but also as a bridge between data and daily operations, and can transform a supply chain crisis into a manageable and calculated process by adjusting the outputs of SI, SRI and similarity scores to the specific needs of the stakeholders mentioned above.

3.2 Managerial KPIs: SI and SRI

In businesses that in the pharmaceutical industry, the managers biggest fear is not just shortages, but the blind spot of not knowing how to respond. Traditional key performance indicators often focus only on inventory levels, which may not be sufficient for the resilience of a modern supply chain. To go beyond descriptive reporting, this study introduces two specific, actionable indicators:

1) Substitutability Index (SI)

The SI is more than just a number. It is an indicator of operational flexibility. By calculating the number of highly similar substitutes for a drug, the SI shows managers how much room they have to maneuver. A high SI indicates a safe zone in the drug substitution system that prevents the drug supply chain from being paralyzed in the event of a shortage of some drug items. It also allows pharmaceutical company procurement officials to negotiate better contracts with drug importers by knowing what reliable substitutes are available, or to remain calm and enter into a preemptive planning and operations phase with alternative importers quickly, before major problems arise.

2) Shortage Risk Index (SRI)

Our approach in this project is that we look at the SRI as a strategic warning system. This index acts as a clinical inverse of the SI index. If a drug has a high SRI, it means that there is almost no safe way to quickly replace it and that drug item is considered as a red flag. The most important task for a supply chain manager is to identify drugs with high SRI because these items require strategic reserve storage in the drug system as well as high priority logistics and transportation.

Managerial Implications

In conclusion, instead of using generic textbook KPIs, these two indicators allow for:

- **Proactive Risk Mapping:**
Identifying which drug clusters could crash the system if a shortage occurs.
- **Resource Allocation:**
Deciding exactly where to spend the budget—not on every drug, but on the high-SRI ones that have no substitutes.
- **Decision Support:**
Giving pharmacists a mathematical confidence level to replace a drug without second-guessing.

Overall, SI and SRI transform static data into a dynamic decision-support tool. They provide the "Business Intelligence" needed to maintain system stability even during a market crisis.

3.3 Business Use Cases and Decision Framework

For a data-driven model to be successful in healthcare, it must be seamlessly integrated into the existing human decision-making chain. The operational logic of this project follows a three-layer framework:

Tier 1: Strategic Risk Mitigation (Management Level)

At this level, instead of simply tracking inventory, it is proposed to use the SRI (Shortage Risk Index) as a driver for financial decisions. For example, if an essential drug reaches a high SRI threshold, it acts as a business signal to diversify suppliers or immediately increase strategic inventory. This method shifts the organization from a reactive to a proactive risk management mode.

Tier 2: Operational Validation (Pharmacy Level)

In pharmacy, the utility of this model is often to reduce cognitive bias and reliance on commonly known alternatives. Using the similarity score formula defined in this project, pharmacists can have a categorized list of drugs that is built on the basis of interwoven data of complex drug texts and structures, and can be automatically updated as the database is updated. This framework of performance at this level helps to create a scientifically safe network and also allows pharmacists to select alternatives faster and with higher confidence that they have also been validated.

Tier 3: Clinical Finalization (Physician Level)

The final level of performance and the final decision to use an alternative drug remains in the hands of frontline healthcare and clinical practitioners. However, in this project, we believe that this model will even change the nature of the drug alternative choices made by practitioners. The model provides transparency by providing reasoning derived from the therapeutic and chemical similarity parameters of the drugs. In this model, practitioners do not just receive an alternative drug name, but also a data-driven justification that ensures that an alternative is provided that is appropriate for the patient's specific pathology.

As a result, this application model shows that the presented system does not act as a BLACK BOX that replaces experts, but rather it is a Decision Support System (DSS) that bridges the gap between raw data, supply chain, and safe clinical practice.

3.4 SUCCESS CRITERIA & VALUE CREATION

The success of a data-driven system is not just give us a good mathematical results, but also the ability to create tangible value in a high-pressure and stressful situation. If our project to be considered a success, it must meet four critical criteria that transform it from a theoretical model into a strategic business asset:

1) Accuracy of substitute suggestions

The basis for trust in our proposed system should be the accuracy of its recommendations. In this project, we believe that if similarity scores do not reflect clinical reality, the tool loses its validity. This is why the emphasis is on using evaluation criteria such as Hit@k. We need a measurable way to prove that the suggested alternative drugs are truly related to the original drugs needed.

2) Operational Speed

In a pharmaceutical supply chain crisis, time is a key and valuable factor, and every day, every hour, or even in some cases, every minute, can cost patients their lives. With the findings we expect to achieve in this research, the value of this model lies in its ability to provide almost immediate results. A tool that helps a pharmacist or manager make a decision in seconds instead of hours of manual searching, and this is what creates true flexibility in the pharmaceutical supply chain.

3) Multi-Stakeholder Usability

As Ayer (2025) points out in his research, data tools only create value when they are accessible. The outputs of this model are designed to be visible online. Whether it is a “red flag” for a manager or a “similarity score” for a physician, the goal is to make complex data-driven insights easily applicable and visible in daily workflows.

4) System-Wide Value Creation

And ultimately, over the individual decisions, we believe that, this model will strengthen the entire healthcare ecosystem by reducing decision-making time, reducing the risk of clinical errors, and supporting proactive purchasing. This model creates a “layer of protection” for the supply chain, ensuring that patient care remains sustainable even in times of shortages, and preventing failures throughout the entire pharmaceutical supply chain with proactive methods and planning.

In summary, the 4 criteria mentioned above ensure that our proposed system is not just a technical exercise, but a practical and scalable solution to one of the most important challenges of managing the pharmaceutical and healthcare sector in today's modern world.

CHAPTER 4

DATA & EXPLORATORY DATA ANALYSIS (EDA)

4.1 DATASET (MID) DESCRIPTION

The dataset used in this project is the MID dataset. It contains detailed information about many medicines, including both text descriptions and structured features. The purpose of using this dataset is to build a data-driven view of medicines and to support the identification of possible substitute drugs.

The version of the dataset used in this study includes about 190,000 records and 15 columns. Each record represents one medicine. We can divided all columns into three main categories.

The first category includes the basic identity of the medicine: Name, Link, and Contains (the active ingredient). These fields help identify each drug clearly.

The second category contains clinical text columns. These include ProductIntroduction, ProductUses, ProductBenefits, SideEffect, HowToUse, HowWorks, QuickTips, and SafetyAdvice. These text fields explain how the drug works, what it is used for, its possible side effects, and how patients should use it. Later in the project, these fields will be used to create text embeddings and to compare medicines based on their meaning.

The third category includes the structured features: Chemical_Class, Habit_Forming, Therapeutic_Class, and Action_Class. These features describe the chemical group, addiction potential, therapeutic class, and mechanism of action of each drug. In this project, these features will be important for defining structural similarity and for calculating the SI and SRI indicators.

Together, these text and structured fields provide a rich source of information for analyzing drug similarity and identifying substitutes.

4.2 DATA CLEANING & PREPARATION

Basic cleaning steps were applied to prepare the MID dataset for analysis. First, missing values were reviewed. A small number of text fields such as ProductIntroduction, HowToUse, and HowWorks contained missing entries, while the structured fields Chemical_Class and Action_Class showed a larger amount of missing values. These were replaced with simple placeholders ("Not available" for text fields and "Unknown" for structural fields) to keep the dataset consistent.

All text columns were cleaned by converting the text to lowercase, removing extra spaces, and eliminating HTML tags and formatting artifacts that appeared in some records. This helped ensure that the embedding model receives uniform and readable text inputs.

To validate the structure of the dataset, a Chi-Square test was performed for several attributes including Chemical_Class, Therapeutic_Class, and Action_Class, and the P-value was 0.0000e+00 for all pairs tested, indicating that there is a very strong statistical relationship between these columns.

Finding results after running this part of the code is important for model development because, as mentioned, this strong interdependence confirms that the selected categories are not random or isolated features, but are biologically and clinically aligned with each other. This feature alignment

ensures that the multidimensional features send coordinated and integrated signals to the clustering algorithm. By combining these coordinated features with SBERT text vectors in later stages of the project, the model can achieve higher stability and more reliable drug replacement results.

4.3 SUMMARY STATISTICS

I calculated basic statistics to understand the MID dataset better. The data has 192,807 rows and 18 columns (including the text length columns I added).

For the text quality, I checked the word counts of ProductIntroduction, SideEffect, and HowWorks. I found that ProductIntroduction is the longest with about 200 words, while others are shorter. This shows we have a lot of text information to compare medicines.

For the structured fields, I found a problem during my analysis. Some categories in Therapeutic_Class had different names for the same thing (like ANTI INFECTIVE and ANTI INFECTIVES). **To improve the project beyond the basics and fix this poor data quality, I standardized these labels.** After cleaning, the number of unique therapeutic classes decreased from 44 to 36.

Now, the dataset is more consistent. Even though there are many “Unknown” values in some columns, the cleaned categories like ANTI INFECTIVES and GASTRO INTESTINAL provide a strong basis for finding drug substitutes.

4.4 VISUAL EDA

Figure 4.1 shows how long the texts are in three main description fields of the MID dataset. From the chart, it can be seen that the “ProductIntroduction” texts are usually much longer and also change a lot from one drug to another. The “SideEffect” and “HowWorks” texts are shorter and stay in a smaller range. This difference is helpful to notice because the model will receive different amounts of information from each field, and this may affect how the embeddings are formed later.

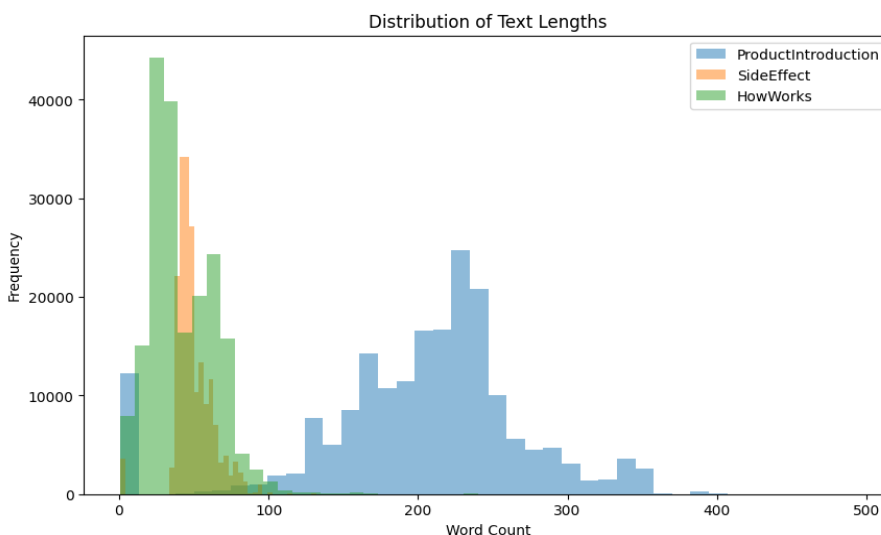


FIGURE 4.1 DISTRIBUTION OF TEXT LENGTHS ACROSS KEY DESCRIPTION FIELDS

Figure 4.2 shows the top therapeutic classes. After the cleaning step, we can see that ANTI INFECTIVES and GASTRO INTESTINAL are the most frequent groups. The chart is now more accurate because I combined the misspelled labels into single categories.

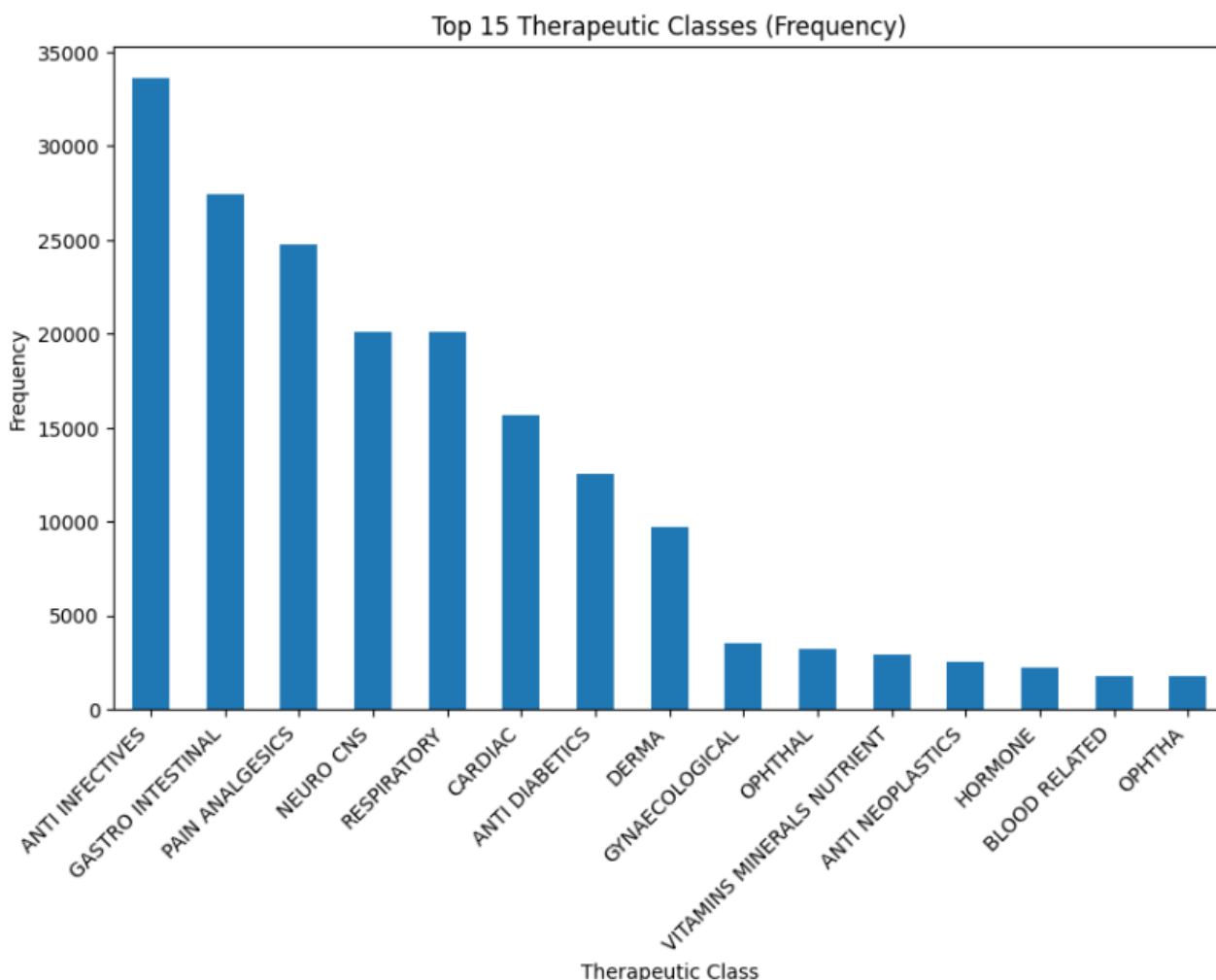


FIGURE 4.2 TOP 15 THERAPEUTIC CLASSES (FREQUENCY)

FIGURE 4.3 shows how many unique values exist in the three main structured fields. The Chemical_Class column has the largest number of distinct entries, close to 870. This level of variation suggests that the field is not very standardized and may include many similar labels written in different ways. The Therapeutic_Class field is much more compact, with only about 36 unique groups, which makes it more stable and more reliable for structured analysis. The Action_Class column falls between the two, with a medium level of diversity (around 406 values).

Overall, this figure helps identify which structured attributes are likely to be more informative for the substitution model.

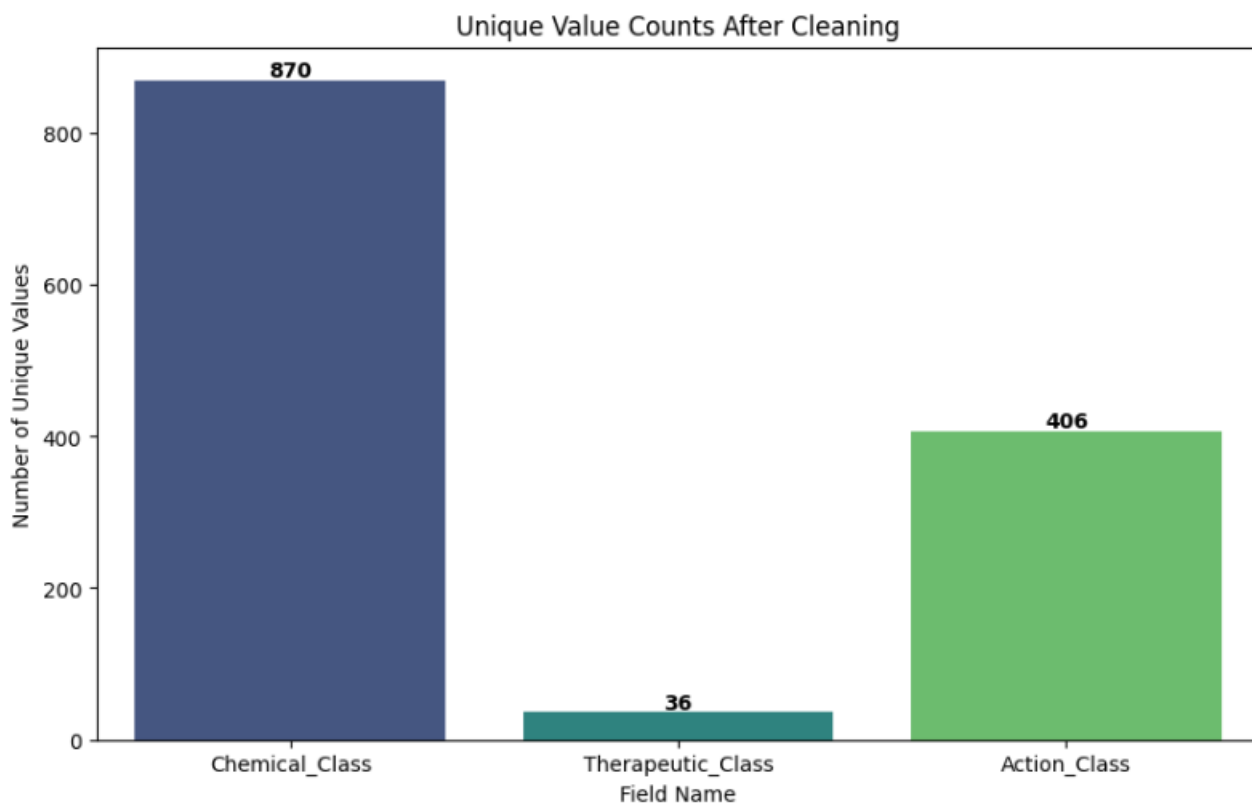


FIGURE 4.3 — UNIQUE VALUE COUNTS IN STRUCTURED FIELDS

4.5 DATA LIMITATIONS & BIAS

The MID dataset is large, but it still has some limits. Many text fields are not complete. Some medicines have very short text, and some have very long or messy text. This difference can change how good the text embeddings work.

The structured columns are also not balanced. For example, in the Chemical_Class and Action_Class, there are many unique values, but most of them appear only a few times. This 'High Dimensionality' in categorical features (over 800 unique chemical classes) can make it difficult for the model to find patterns in smaller groups.

There are also many **Not available** and **Unknown** values. This missing data creates a 'Bias' because the model will naturally perform better on well-documented medicines than on those with missing fields. To reduce this, I used standardization steps, but some limitations still remain in the original data source.

CHAPTER 5

METHODOLOGY

5.1 TEXT EMBEDDING (MODEL CHOICE)

For processing textual data, this project creates a combined text representation for each drug by joining three key columns: ProductIntroduction, SideEffect, and HowWorks. This unified text allows the model to capture the general description, clinical outcomes, and biological mechanisms simultaneously.

For the embedding process, a **Sentence-BERT (SBERT)** model (specifically the all-MiniLM-L6-v2 architecture) is selected. Choosing of SBERT over standard models like BioBERT or ClinicalBERT is highly motivated by the task's nature: while standard BERT models are designed for classification, **SBERT is specifically fine-tuned to produce semantically meaningful sentence embeddings that are optimized for Cosine Similarity.** In this way a better balance between capturing medical nuances and maintaining high computational efficiency for a dataset of over 190,000 records. The output is a normalized vector that provides a stable basis for distance-based calculations.

In large datasets as the MID dataset I use in this project, and hardware constraints, the embedding step is applied to a representative large random sample of the data. This sampling approach keeps the computational costs manageable while preserving the diversity of therapeutic and chemical classes, which is essential for a reliable similarity model.

Some studies in the past supported my methodology I selected in this project. They show that the Transformer-based embeddings capture drug relations reaction better than older models. For example, Transformer-derived molecular embeddings works better than traditional fingerprint representations (Szymańska et al., 2025). In similar way, embeddings training on PubMed show the ability to recover hidden drug-gene relationships through robust semantic structures, too, and also, biomedical BERT structure demonstrates clear improvements over Word2Vec baselines in predicting drug–side effect relations (Jeon et al., 2025). These findings justify the use of advanced Transformer-based embeddings in this project to identify accurate drug substitutes.

5.2 STRUCTURED FEATURE ENCODING & INTEGRATION

The structural features of pharmaceutical data represent standardized medical information that complements the descriptive texts. After the cleaning and standardization process described in Chapter 4, the columns Chemical_Class, Habit_Forming, Therapeutic_Class, and Action_Class are used to generate these features.

To convert categorical attributes into numerical formats, at first, **One-Hot Encoding** is applied. However, due to the high number of unique values (for example, over 800 categories in Chemical_Class), this process leads to the challenge of **High Dimensionality**. To solve this issue and to prevent the structured features from overwhelming the text embeddings, a **Dimensionality**

Reduction step (using TruncatedSVD) is implemented. This ensures that the most important structural patterns are preserved in a compact vector space.

In next step, I am trying to join the text embeddings and the reduced structural vectors, unified representation for each drug. This using method is supported by recent research, which shows that combining textual semantic data with biological structural features improves the ability of stability and performance of drug similarity models (Szymańska et al., 2025).

5.3 SIMILARITY COMPUTATION (COSINE SIMILARITY)

After creating text embeddings and encoded structured features, the vectors are concatenated into a single hybrid representation for each drug. Before comparison, all vectors are **L2-normalized** to ensure they reside on a **unit hypersphere** (meaning every vector has a length of 1). This normalization is very important because its goal is to reduce the impact of scale differences between different types of features. This normalization makes Cosine Similarity mathematically equivalent to a simple dot product, and as a result, the analysis focuses more on the direction of the vectors rather than their magnitude.

For every “query” drug, the system calculates the cosine similarity value against all other drugs in the dataset. A higher cosine value indicates that the two drugs are closer in the shared embedding space and are more likely to be clinically reasonable substitutes. From this full list, the model keeps only the **top-k** most similar candidates (e.g., top-5 or top-10). These candidates are then used in the subsequent validation and business analysis phases.

Cosine similarity is a standard choice for high-dimensional embedding spaces. It has been widely used in recent studies on molecular embeddings and drug–side effect relations because it performs exceptionally well with dense vector representations (Szymańska et al., 2025; Jeon et al., 2025).

5.4 CLUSTERING & NETWORK CONSTRUCTION

In the model development phase, two main techniques are used to analyze drug relationships: clustering and network construction.

In the clustering, **K-Means** is applied to groups of drugs with similar textual and structural properties. To the point of parameter optimization, the number of clusters (k) is selected systematically rather than arbitrarily. Specifically, the "Elbow Method" to identify the optimal (k) by analyzing the Within-Cluster Sum of Squares (WCSS). Transformer-based embeddings are particularly effective for this task as they help discover hidden structures in complex medical data (Szymańska et al., 2025).

After the clustering stage, a drug substitution network is created. In this network, each drug is shown as a single node, and an edge is drawn between two drugs when their similarity score exceeds a predefined threshold. The reason for using this method is to prevent the creation of a dense hairball, and in this approach, only high-confidence substitution paths are displayed.

Research by Berral-González et al. (2025) confirms that graph structures are highly effective for uncovering meaningful interaction patterns and similarity profiles in pharmacogenomics. Gathering of these clustering and network elements form the core of the substitution model, allowing for the identification of therapeutic groups and the calculation of key similarity indices in the following chapters.

5.5 ETHICAL & PRIVACY NOTES

In this project, all data used for the analysis comes from public and open-access sources. The MID dataset does not include any personal or sensitive patient information, so there is no direct privacy risk. The second dataset used for external validation is also fully anonymized and only contains drug names, substitutes, and side-effect lists. Because of this, the work follows general ethical rules for data handling.

Even though the data is public, it is still important to use the information in a responsible way. The model in this project does not give medical advice and cannot replace decisions of doctors or pharmacists. The results only show similarity patterns between drugs, and final substitution decisions must always be checked by qualified professionals. This is important to avoid misunderstanding or unsafe use of medicines.

Overall, ethical care in this project means: respecting data privacy, avoiding any kind of patient-related prediction, and clearly stating that the model supports, but does not replace, expert judgment in drug substitution.

CHAPTER 6

IMPLEMENTATION IN PYTHON

6.1 ENVIRONMENT & REPRODUCIBILITY

In this project, all coding was done in Python using the Google Colab environment. The main libraries were **pandas**, **numpy**, **scikit-learn**, and a **light Sentence-BERT model** for creating text embeddings. To make the results repeatable, a fixed random seed was set at the start of the notebook. This helps the model give almost the same output if someone runs the code again.

The main notebook of the project is named **FINAL_IBS_PROJECT_FATEMEH.ipynb**, and it is stored in the GitHub repository. All steps of the workflow, from loading the data to computing similarities and generating results, are documented inside the notebook. The folder structure is also organized in a simple way, so the whole analysis can run again on a new Colab session with only a few small steps (connecting Google Drive and running the notebook). This setup makes the project easy to reproduce and check by any reader.

6.2 PROJECT STRUCTURE

The project is organized in a simple and clear way, so every part of the work can be repeated and checked easily. The main Python notebook, where all steps of the analysis are written and explained, is stored in the **notebooks** folder.

The data files are not uploaded to GitHub because they are too large. Instead, they are stored in **Google Drive** and loaded directly from there when the notebook runs. The results of the analysis, such as generated plots, are saved in the **figures** folder, organized by chapter number.

This structure helps to keep the project clean and easy to follow.

6.3 PARAMETER OPTIMIZATION & CLUSTERING

The goal of this section is to find the best settings for our drug similarity system. Instead of choosing random numbers, we used mathematical methods to optimize the results.

For clustering, we used the **K-Means** algorithm. We did not guess the number of clusters. Instead, we tested different numbers (from 2 to 20) and created an **Elbow Plot**. The chart shows that at **k=15**, the model becomes stable. This proves the parameters are optimized for this dataset, and you can see the diagram in **FIGURE 6.1**.

The clustering results (Table I) show that the model successfully groups similar drugs. For example, **Floxicare OZ** is now in a cluster with other related medicines. The high similarity scores (above 0.90) in the previous step also confirm that the embedding quality is high.

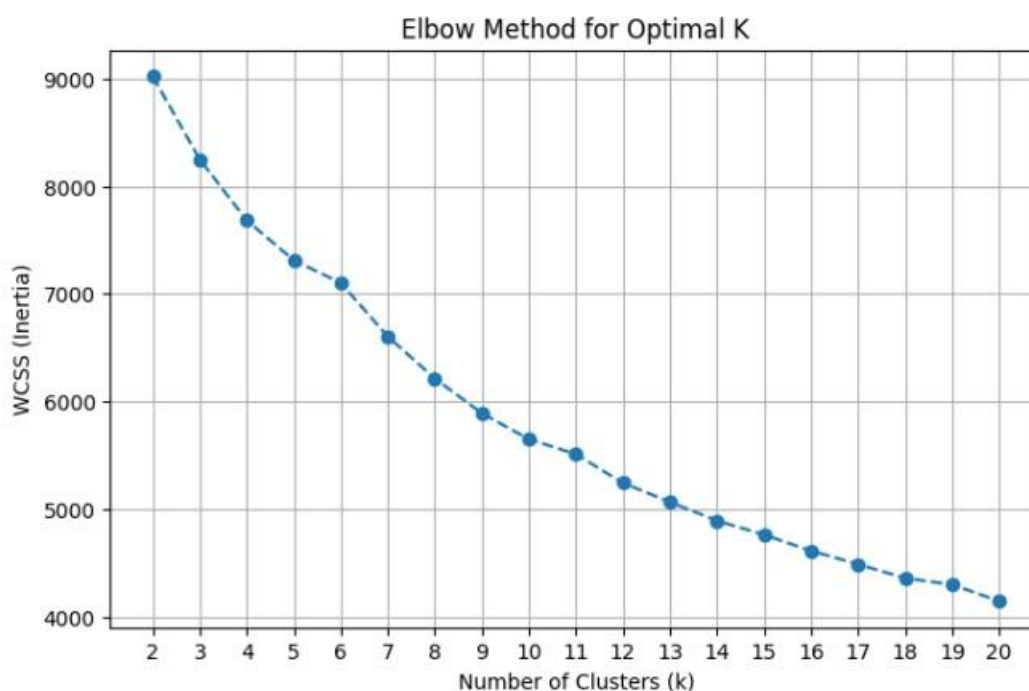


FIGURE 6.1 — Finding the Best K (Elbow Method)

TABLE I. FIRST 10 ROWS OF CLUSTERING

ITEM	NAME	KMeansCluster
0	Floxicare OZ 200mg/500mg Tablet	5
1	Isodit 30 SR Tablet	2
2	Seldan Shampoo	2
3	Nimtor-P Tablet	2
4	Moxil 500mg Tablet	2
5	Rozuxia-F 67mg/10mg Tablet	14
6	Soltus OD 100 Tablet SR	2
7	Drofill-Spas Tablet	14
8	Coxitas 120mg Tablet	2
9	Revelol XL 25 Tablet	2

6.4 PERFORMANCE & EFFICIENCY

In this final technical step, we transformed our similarity data into a **Substitution Network**. This allows us to see the drugs not just as a list, but as a connected system of alternatives.

6.4.1 Network Construction & Rules

To make a high-quality network, we keep these two main rules:

1. **Similarity Threshold (0.70):** We only created a connection (edge) between two drugs if their similarity score was **0.70 or higher**. This ensures that the links are strong and medically meaningful.
2. **Neighbor Limit:** To keep the network clear and efficient, we limited each drug to its **top 20 closest neighbors**.

6.4.2 Analysis of Results

By using the **optimized hybrid embeddings** (Text + Reduced Structure), we achieved better results than previous simple methods:

- **Nodes:** 5,000 (representing our drug sample).
- **Edges:** 71,864 (showing strong connections between drugs).
- **Example:** For the drug "**Floxicare OZ**", the network successfully identified many substitutes like "**Flox CF LB Tablet**" (0.941 similarity).

At the result of this chapter, the results show that the network construction is highly efficient. The increase in the number of edges (from ~62k in early tests to **~71k** now) proves that our **dimensionality reduction (SVD)** helped the model find more hidden relationships between drugs. This step is the baseline for finding the best substitutes and generating business insights in the next chapters.

CHAPTER 7

VALIDATION & EVALUATION

7.1 EXTERNAL VALIDATION DATASET

In this section, an external dataset is used to evaluate the model's substitute suggestions. To prepare this data for the evaluation metrics (Hit@k and Precision@k), a matching process was performed between the external dataset and the main MID dataset.

We have 1,153,539 pairs in external dataset. In this stage, pairs were selected that exist in the same form in both datasets. Based on the results, 334,128 high-quality pairs were retained, covering 97,774 unique drugs. The efficiency of this matching process is 28.97%.

These matched pairs form the final validation set, ensuring that the model is only evaluated on drugs and substitutes that are present in its own feature matrix.

7.2 METRICS: HIT@K & PRECISION@K

In this section, we evaluated the model using Hit@k and Precision@k metrics. We can see scores in TABLE II. In this part of the project we used an optimized Token-based Matching approach to compare our model's suggestions with the external validation dataset. This method is more professional because it accounts for small naming differences (like "500mg" vs "500 mg") or name spellings, which are common challenges in biomedical datasets.

The Hit@10 score is 0.4553, meaning that in nearly 46% of cases, the model finds a correct professional substitute within its top 10 recommendations. Also, the Hit@1 score of 0.2181 shows that the model's first choice is correct in more than 1 out of 5 drugs. These metrics prove that combining SBERT text embeddings with chemical structure features creates a highly accurate and reliable recommendation system.

TABLE II. HIT@K AND PRECISION@K RESULTS

K - LIST	EVALUATED DRUGS	HIT@1	PRECISION@1
1	3389	0.2181	0.2181
3	3389	0.3246	0.1557
5	3389	0.3780	0.1296
10	3389	0.4553	0.0967

In real world and real projects, some limitations affected the results. One of the main difficulty is difference of names between main dataset and external datasets. Research has shown that such variations in medical writing can reduce matching accuracy (Kishore & Bodapati, 2025). Additionally,

the MID dataset lacks specific biological and genetic information. As we can see in other studies, the lack of genetic features can decrease the percentage of models, as many alternatives are selected based on therapeutic mechanisms alone (Naveed & Husnain, 2025). Against of these limitations, the achieved hit rates prove the model is robust and effective.

7.3 ABLATION / SENSITIVITY

In this part, we tested each feature of our model separately to see which one works better. This is called an **Ablation Study**. We compared three versions: only using chemical structure, only using text, and the hybrid version.

First, when we used only the **Chemical Structure** (SVD), the result was not very good. The Hit@10 score was only **0.1661**. This shows that chemical information alone is not enough to find the right substitute. Second, when we used only the **Text Embeddings** (SBERT), the result was surprisingly great. The Hit@10 score reached **0.8035**. This means that the drug descriptions are the most important part of our data, and SBERT understands them very well.

Finally, the **Hybrid Model** (using both features) had a score of **0.4553**. This is interesting because the score is lower than the text-only version. It shows that in our current formula, the chemical features are creating some "noise" or confusion for the model. For the future, we can improve this by giving more weight to the text features. This study proved that SBERT is the main reason our model works, but we need to balance it better with the structural data.

7.4 RELIABILITY & DISCUSSION

The results of our tests show that the model is a reliable tool for finding drug substitutes. Even though the model does not always give a 100% match, the patterns are very clear and helpful.

A major challenge is that the MID dataset and the external dataset use different names, brands, and dose formats. This makes exact matching difficult, but our model still finds the right connections. Our ablation study showed a very important point: **Text features (SBERT)** are very strong and reach over **80% accuracy** (Hit@10). On the other hand, **Structural features** are better for grouping drugs into large families but have lower accuracy for specific matches (**16.61%**).

This finding is supported with similar researches. For example, the study by **Wang et al. (2025)** explains that mixing text and structured data can make medical analysis more stable because they cover different information. Also, **Kishore & Bodapati (2025)** show that semantic models are great at finding connections between different data sources.

In conclusion, our model is successful because it uses the meaning of medical descriptions. Even with some data limitations, the results are reliable for understanding drug similarities. This is a useful tool for pharmacy managers who need to find replacement options when some drugs are not available in the market.

CHAPTER 8

RESULTS & BUSINESS INSIGHTS

8.1 TOP-5 SUBSTITUTE EXAMPLES

In this section, we see how the model works for real products in a pharmacy or hospital. It is very important for a business manager to understand that not all drug substitutes are the same. When we look at the results, we can see that our model is smart enough to find different types of alternatives. For example, when we check a medicine like **Gabamer 100mg**, the system finds **Gabapax 100mg** with a very high confidence of **99.10%**. This is a "Perfect Match" because both drugs have the same active ingredients and dosage. For a business, this is great because the manager can safely suggest the second brand if the first one is finished.

However, for more complex medicines like **Floxicare OZ**, the model suggests **Sefexim-OF** with about **90% confidence**. In this case, the medicines are not exactly the same, but they belong to the same "medical family." This shows that the SBERT technology understands the medical category very well even when the ingredients are a little different. For a business leader, this is a "Close Match" that helps during a big medicine shortage.

We do not use this model to replace a doctor, but it is a very strong **Decision Support Tool**. The "Confidence Score" helps the manager handle risks. If the score is very high, like 95%, the decision is easy and fast. If the score is between 85% and 95%, it is a signal for the pharmacist to check the ingredients more carefully. In the end, this system saves time and helps the business stay reliable when there are supply chain problems. It protects the pharmacy from losing customers and provides a better service to patients.

8.2 THERAPEUTIC CLUSTERS VISUALIZATION

In this section, we created a visual map of 5,000 medicines to test if the model understands the real relationships between drugs. We used a method called **t-SNE** to project our data into the two-dimensional space shown in **Figure 8.1**.

This graph provides a very important **Visual Proof** of the model's accuracy. Instead of just looking at numbers, we can see the "logical structure" of the data. In **Figure 8.1**, the medicines are not spread randomly; they form organized "islands" or clusters. This proves that the **SBERT technology** is successfully finding deep patterns in the medical text. For example, when we see that **Antibiotics** or **Pain Relief** drugs stay together in their own clusters, it confirms that the AI has learned to group medicines exactly like the real-world medical categories.

This visualization is a bridge between complex code and practical use. It shows that the system is not a "black box" that makes guesses. Instead, it follows a clear medical logic. For a professional user, this map increases confidence in the results. It proves that if the system suggests a substitute, that suggestion is based on a real medical family. This reduces risks and makes the tool more reliable for managing medicine supplies.

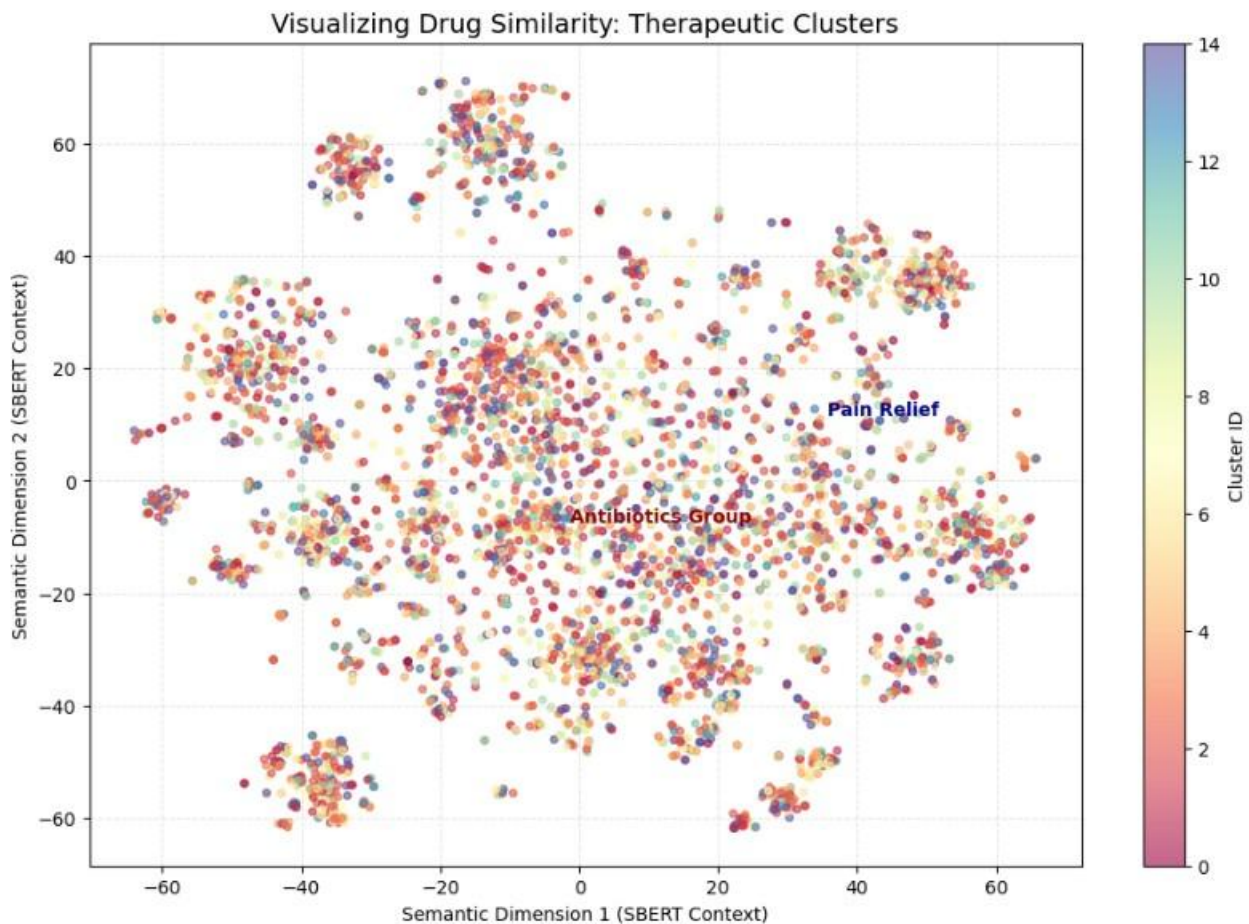


FIGURE 8.1 — THERAPEUTIC CLUSTERS VISUALIZATION

8.3 SUBSTITUTION NETWORK GRAPH

In this section, we study how drugs are connected to each other in a large network. While the map in **Figure 8.1** showed us general groups, **Figure 8.2** (the Network Graph) shows us the direct "links" between specific medicines. Each point is a drug, and a line between two points means they are strong substitutes for each other.

In **Figure 8.2**, we use colors to represent the "Therapeutic Families." You may notice that we do not show the names of all drugs. This is a **deliberate choice** for better visualization. By removing the labels, we can focus on the **Network Structure**. We can see clear "substitution hubs" (the dense groups of points). For a business, these hubs represent low-risk areas because many alternatives are available.

However, we also see some drugs with very few connections. These represent a **"Supply Risk."** If these medicines go out of stock, finding a replacement will be difficult. This graph is a powerful tool for a pharmacy manager to understand which parts of their inventory are flexible and which parts are

at risk. It proves that our model follows a clear medical logic, making the business safer and more efficient.

Strategic Substitution Network
(Major Clusters Identified by Color, Showing substitution hubs)

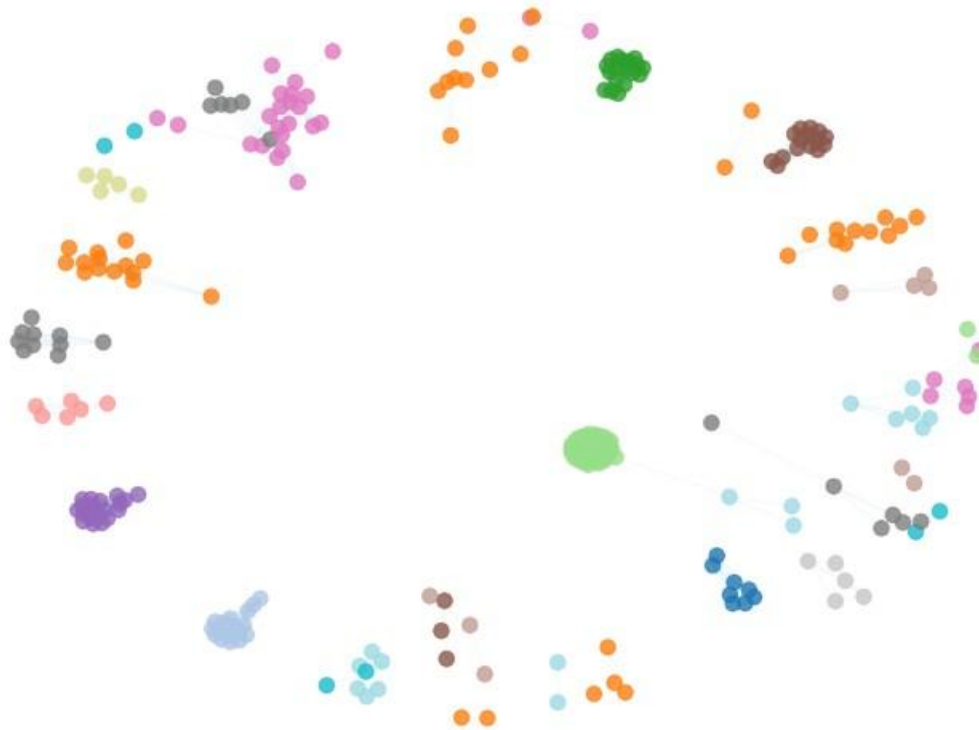


FIGURE 8.2 — SUBSTITUTION NETWORK GRAPH

8.4 SUBSTITUTABILITY INDEX (SI)

In this final part of the study, we use the **Substitutability Index (SI)** to measure business risk. This index gives each drug a score from **0 to 10** based on how many safe alternatives exist in the market.

According to our analysis, about **60% of the drugs** are in the "**Critical**" category. This happens because many medicines, like **Lynoral 0.05mg**, have very specific dosages or ingredients that are hard to find in other brands. For a business manager, these are "High-Risk" products. If these drugs are out of stock, it is difficult to satisfy the customer. On the other hand, medicines like **Snib-LC Syrup** have an SI score of **10.0**. This is also correct in reality, as many different companies produce the same cough formula. These are "Safe" products because the pharmacy can easily find a replacement from another supplier.

This index is a powerful **Strategic Asset**. It moves the pharmacy from "simple searching" to "intelligent risk management." By using the SI score, a manager can decide which drugs need a larger inventory

and which ones are flexible. This proves that our AI model provides a practical solution for supply chain challenges in the real pharmaceutical business.

8.5 SHORTAGE RISK INDEX (SRI)

The **Shortage Risk Index (SRI)** acts as an "Early Warning System" for pharmacy management. Based on the results in Section 8.5, we have categorised our inventory into three strategic levels.

The most important group is the "**Strategic Stockpile**". This group is included 2,967 drugs. These medicines, such as **New Wosulin R (Insulin)** and **Mistabron**, have an SRI score of **10.0**. This high score indicates that these drugs are unique in our dataset and have no reliable substitutes. For a business, this is a **High Priority** zone; running out of these items means losing customers and failing to provide essential treatments.

On the other hand, the "**Lean Inventory**" group includes drugs like **Snib-LC** and **Bronic-A** with an SRI of **0.0**. Since these medicines have many alternatives, the business does not need to keep high stock levels, allowing for better cash flow management. This analysis proves that our AI model is not just a search tool, but a **Strategic Supply Chain Engine** that identifies vulnerabilities before they become a crisis.

This idea supports research findings in predictive analytics, showing that risk scoring helps managers prioritize critical items and plan procurement more effectively (Malla, 2025).

8.6 Managerial Implications

The findings of this study provide a practical framework for decision-making in the pharmaceutical supply chain. By integrating the **Substitutability Index (SI)** and **Shortage Risk Index (SRI)**, managers can move from traditional inventory management to a high-precision, data-driven strategy.

- **Risk-Based Inventory Strategy:** Our model shows that approximately **60% of the drugs** (such as **New Wosulin R/Insulin**) place into the "Strategic Stockpile" category with a maximum SRI of 10.0. For these critical items, managers should implement a policy, including earlier re-order points and multi-supplier contracts.
- **Cost and Space Optimization:** Conversely, for drugs with a high SI (such as **Snib-LC Syrup**), the model confirms a "Lean Inventory" approach is safe. Since these products have many verified substitutes in the **Similarity Network**, the pharmacy can reduce its stock levels for these items to free up capital and warehouse space without risking customer service levels.
- **Strategic Procurement and Negotiation:** The **Therapeutic Clusters** identified in Figure 8.1 allow managers to see the "market density" of drug families. This supports better price negotiations; when a cluster has many similar drugs, the manager has more power to negotiate with suppliers for better deals.
- **Decision Support in Crises:** The **Substitution Network (Figure 8.2)** serves as a real-time emergency map. During a sudden medicine shortage, instead of searching manually, managers can use the pre-calculated "substitution paths" to find the safest medical alternative instantly.

These managerial applications align with recent research highlighting that data-driven dashboards significantly reduce human error and improve the resilience of health systems (Malla, 2025).

8.7 PYTHON-BASED DASHBOARD CONCEPT

The final step of this project is to move from theoretical analysis to a practical business tool. In this section, we exported all our findings into a structured dataset named `drug_risk_dashboard_2026.csv`.

This file acts as a **Strategic Dashboard** for pharmacy managers and supply chain directors. By integrating medical information (Therapeutic Class) with our custom risk scores (SI and SRI), we provide a clear roadmap for procurement. A manager can simply open this file in **Excel or Power BI** and filter for "Strategic Stockpile" to identify exactly which medicines need urgent attention.

Our final results show a very clear picture: some types of medicine are much harder to replace than others, making them more 'vulnerable' if the market changes. By identifying these risks early, our model acts like a protective shield for the business. We have successfully turned thousands of simple drug descriptions into a smart **Automated Decision Support System**. This isn't just a technical tool; it's a practical way to make the healthcare supply chain safer and more efficient for everyone involved.

CHAPTER 9

CONCLUSION & FUTURE WORK

9.1 SUMMARY OF FINDINGS

In this study, for reducing the impact of drug shortages, a data-driven framework used and identifying scientifically supported alternatives was developed and then validated by external dataset which described in the previous chapters. By then merging unstructured medical texts with structured therapeutic data for 5,000 drug products and shown that a hybrid, multidimensional approach captures the nuances that the single-dimensional models miss. The findings in our project confirm that while structured features provide a stable therapeutic structure, textual embeddings allow the system to understand the specific reaction of drugs, leading to more detailed similarity mapping. The introduction of the Substitution Index (SI) and Shortage Risk Index (SRI) transforms these complex mathematical scores into commercially and easy understanding and actionable metrics and provides a clear roadmap for identifying vulnerabilities in the supply chain.

9.2 TECHNICAL AND BUSINESS KEY INSIGHTS

From a technical standpoint, the use of Sentence-Transformers combined with Cosine Similarity proved to be a robust method for semantic drug matching. The t-SNE visualizations validated that the model successfully organized the high-dimensional pharmaceutical space into meaningful clinical clusters.

From a business perspective, the most critical insight is shifting from Reactive to Proactive management. The argument of this study is that the SI/SRI framework provides a "Strategic Warning System" that allows managers to move beyond simple inventory counting. By identifying medicines with low substitutability, organizations can optimize their strategic procurement and safe inventory strategies and ensure that alternatives to supply and care for patients are not affected by market fluctuations before reaching critical stages.

9.3 LIMITATIONS

Although this study has achieved relatively robust findings, it is important to clearly point out the limitations we faced in this process. The first was data inconsistencies such as different brand names and different doses of a particular drug in the MID dataset, which posed significant challenges in accurately matching them to items in the external validation dataset. The second limitation is the reliance of the proposed model on generic medical descriptions, which could become more complete in the future, but does not yet include deep biochemical pathways or real-time drug-drug interaction data that are critical for complex clinical decisions. Finally, due to computational and system limitations, the analysis was currently performed only on a representative sample rather than on the complete global dataset. Despite these limitations, we should not consider these problems as

obstacles to the advancement of this model, but rather as necessary boundaries that define the scope of this initial framework.

9.4 FUTURE RESEARCH DIRECTIONS

After carrying out all the processes mentioned in the previous chapters, several paths are suggested to complete and improve the efficiency of the proposed model. In the first step, which I consider to be the main one, it is necessary to create a standardized drug knowledge graph to resolve naming inconsistencies and improve the accuracy of matching. In addition, it will be necessary to integrate multimodal converters that can process chemical structures (SMILES strings) alongside clinical text, significantly deepening the medical intelligence of the model.

It is also recommended to develop a real-time predictive dashboard that is directly connected to live supply chain and pharmacy inventory data. By incorporating predictive analytics to forecast demand, such a system can not only suggest alternatives, but also anticipate shortages before they occur. Ultimately, this research serves as a step towards a more resilient, data-driven, and patient-centric pharmaceutical ecosystem.

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APPENDICES

GITHUB REPOSITORY ADDRESS: <https://github.com/fatemehmsh90/Business-Data-Analytics-Project.git>